PoseFit AI

AI-Powered Gym Assistant

Presented by :
Artem Shkurat (SHK22612576)

Supervisor:
Wei Li

Model:
Final-Year Project (CMP-X301-0)

Problem Statement

Problem 1

Beginner gym users often struggle to identify different exercises, understand which muscle groups they target, and learn how exercises should be performed correctly. This can make the gym environment confusing and reduce user confidence during workouts.

Problem 2

Gym users often struggle to remember workout information such as sets, repetitions, weights, and personal records. Without a structured system, tracking long-term fitness progress becomes inconvenient and inconsistent.

Problem 3

Many users perform exercises with incorrect posture or movement technique, especially when training without supervision. Without proper guidance, users may develop poor exercise habits, reduce workout effectiveness, or increase the risk of injury.

Aim & Objectives

Aim

To develop an AI-powered mobile fitness application capable of recognising gym exercises in real time using smartphone camera processing and machine learning technologies.

Objectives

1. Develop a mobile application using Flutter and Dart.
2. Implement real-time exercise recognition using MediaPipe and TensorFlow.
3. Provide exercise information, technique guidance, and targeted muscle group details.
4. Create a workout logging and Personal Records tracking system.
5. Implement secure user authentication and account management features.
6. Evaluate the usability, accuracy, and reliability of the application.

Key Application Features

AI Recognition

Real-time pose estimation powered by MediaPipe and TensorFlow. The app identifies exercises through your camera, displaying exercise name, target muscles, and technique tips instantly.

Data Privacy

GDPR-compliant data handling with secure authentication and user-controlled camera permissions. No video recordings are stored, and all exercise recognition is processed locally on the device to improve user privacy and security.

Workout Logging

Track sets, reps, weights, and Personal Records with a built-in workout logging system designed for long-term fitness progress.

User Authentication

Secure account system with login, registration, password recovery, and protected user data storage. Each user can securely manage personal workouts and records.

Technologies Used

Flutter

Cross-platform SDK used to develop the mobile application interface and navigation system.

Dart

Primary programming language used for developing the Flutter mobile application and application logic.

MediaPipe

Real-time pose estimation framework used to detect body landmarks and track user movements.

TensorFlow

Machine learning framework used for exercise prediction and movement classification.

MySQL Database

Used to store user accounts, workout logs, exercise information, and machine learning datasets.

Python

Used to develop the backend server, machine learning workflows, and communication between the mobile app and database.

Backend API

Handles communication between the Flutter application and the MySQL database through server-side requests.

Exercise Recognition Workflow

1. Camera Consent
The application requests user permission before accessing the smartphone camera.

2. Camera Input
Real-time camera frames are captured and processed continuously during exercise.

3. Pose Detection
MediaPipe detects body landmarks and tracks user movement positions in real time.

4. Angle Calculation
Joint angles and movement measurements are calculated using landmark coordinates.

5. Exercise Classification
TensorFlow analyses movement patterns and predicts the performed exercise.

6. Confidence Check
The system evaluates prediction confidence to reduce inaccurate exercise recognition results.

7. Result & Exercise Display
The application displays exercise information, target muscles, and technique guidance.

TESTING & EVALUATION

Functional Testing

Core application features including authentication, camera streaming, exercise recognition, workout logging, and Personal Records were successfully tested during development.

Recognition Accuracy

Exercise recognition performed best when body positioning, lighting conditions, and movement execution were correct. Recognition accuracy occasionally decreased during fast or unclear movements.

Usability Testing

Several users tested the application by performing exercises and interacting with the system. Feedback indicated that the application was simple to navigate and useful for workout tracking.

Successful Features

- Sign up / Login
- Camera
- Workout logs
- PR system

Limitations

- Lighting sensitivity
- Small dataset

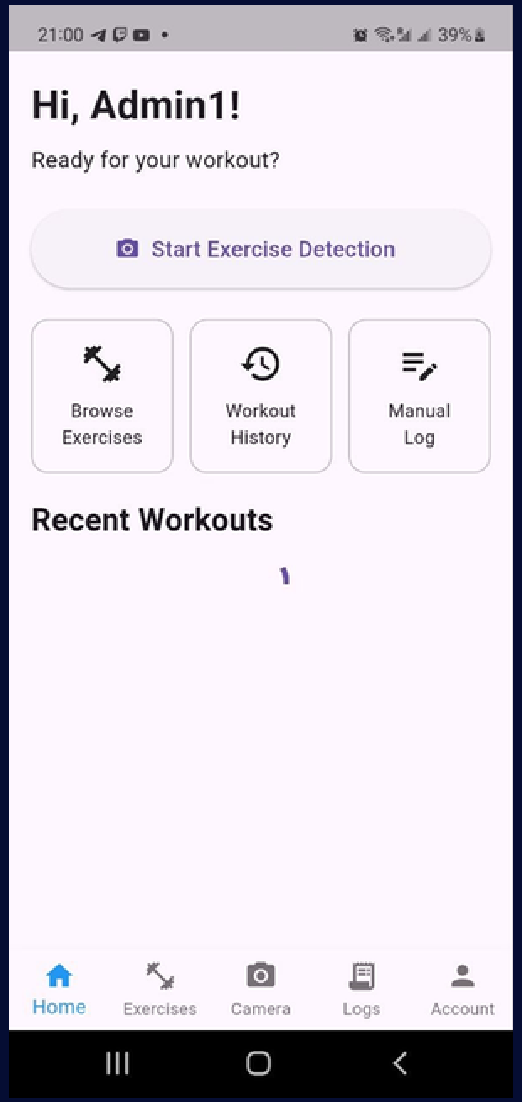
User Feedback

- Easy navigation
- Helpful exercise info
- Useful logging

CHALLENGES FACED

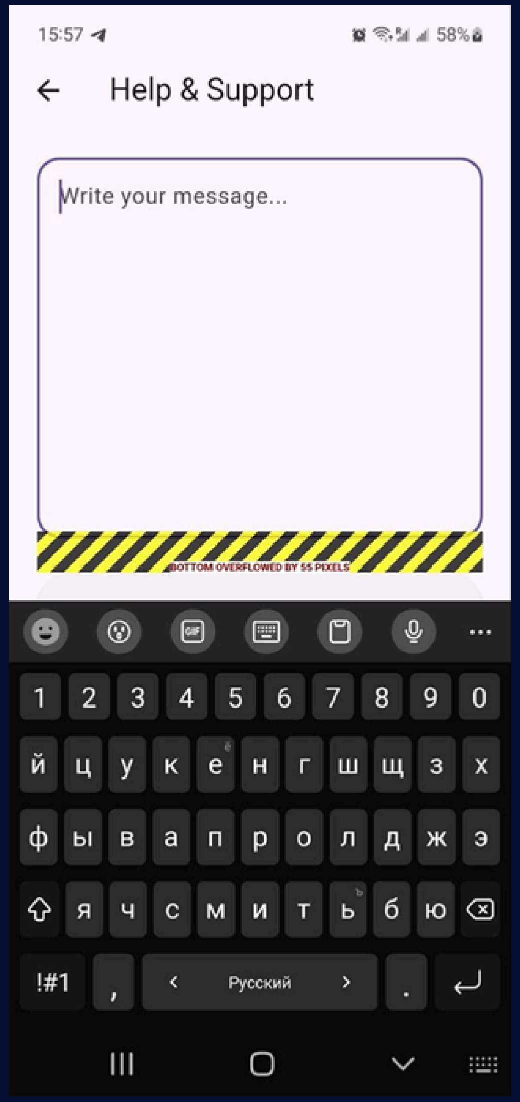
Backend Connectivity

Backend communication occasionally failed due to local network configuration issues and changing server IP addresses during development.



User Interface Stability

Several UI issues such as keyboard overflow, loading freezes, and widget dependency errors required repeated debugging and interface refinement.



Camera Permission Handling

Managing camera permissions across different application states caused instability and required additional permission validation and lifecycle handling.

Exercise Recognition Accuracy

Recognition accuracy varied depending on lighting conditions, body positioning, movement speed, and limited training data availability.

Current State

1. Basic exercise dataset with limited exercise variety for recognition and classification.

2. Manual workout logging requiring users to input sets, reps, and weights by hand.

3. Android-only support with no current compatibility for iOS devices.

Future Plans

1. Expanded exercise dataset covering comprehensive exercises types and variations.

2. Automatic data collection with AI-powered rep counting and weight detection.

3. Cross-platform support for both Android and iOS devices to improve accessibility and compatibility.

CONCLUSION

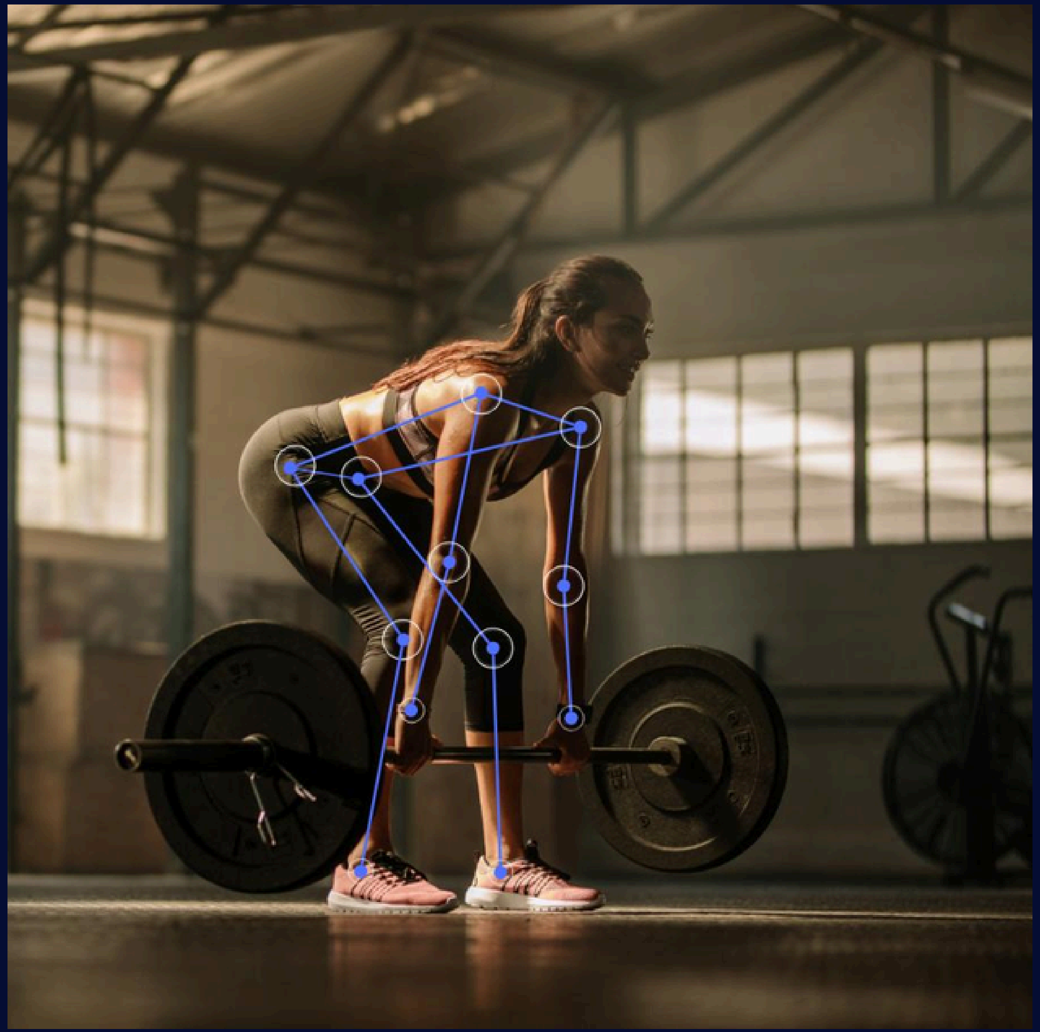
• The project provided valuable experience in mobile application development, backend integration, database management, and machine learning implementation.

• The development process improved problem-solving, debugging, and system design skills through continuous testing and refinement.

• Working with real-time camera processing, MediaPipe, and TensorFlow strengthened practical understanding of computer vision and AI technologies.

• The project also improved knowledge of software engineering practices, including authentication systems, usability evaluation, and application architecture design.

• Overall, the development of PoseFit AI significantly improved both technical knowledge and practical software development experience.



PoseFit AI

AI-Powered Fitness for Everyone

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Thank You

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